Renal Physiology

Renal Water Handling

Frank Petrassi, PhD
Office 348C
fpetrassi@icom.edu
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Learning Objectives

1. Contrast the anterior hypothalamic osmoreceptor-ADH feedback system regulating body fluid osmolarity with baroreceptor-mediated pathways governing extracellular fluid volume.

2. Explain how countercurrent multiplication in the loop of Henle, urea recycling via inner medullary collecting ducts, and countercurrent exchange in the vasa recta establish and preserve the hyperosmotic corticomedullary gradient.

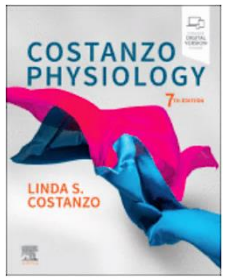
3. Detail the V_2 receptor-mediated cellular signaling cascade in collecting duct principal cells that directs aquaporin-2 trafficking and regulates luminal water permeability.

4. Trace tubular fluid osmolarity changes along the nephron during states of water deprivation versus water diuresis, and calculate osmolar and free-water clearance.

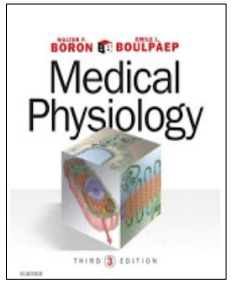
5. Differentiate the etiologies of central diabetes insipidus, nephrogenic diabetes insipidus, and SIADH.

Resources

READ



Chapter 6



Chapter 36, 38, 40

DO

* **TopHat Questions**
- **Guyton and Hall Review Questions (Library)**
- **Costanzo Cases**

TOP HAT

Join Code:

Water Balance Is Not Sodium Balance

Total body Na^+ content is the major determinant of ECF volume.

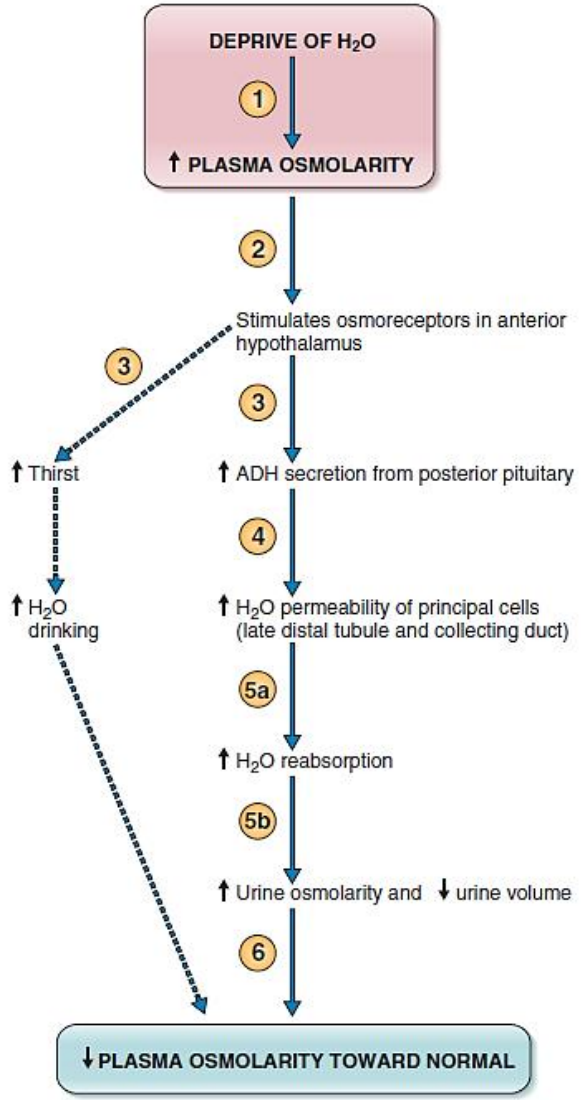
Body water relative to total body solute determines plasma osmolarity and plasma [Na^+].

- Normal plasma osmolarity ≈ 290 mOsm/L.
  - Often simplified to ≈ 300 mOsm/L.
- Urine osmolarity can vary from approximately 50 to 1200 mOsm/L.
- Final regulation of water excretion occurs primarily in the late distal nephron and collecting ducts.

Osmoregulation: Response to Water Deprivation

- Water deprivation:

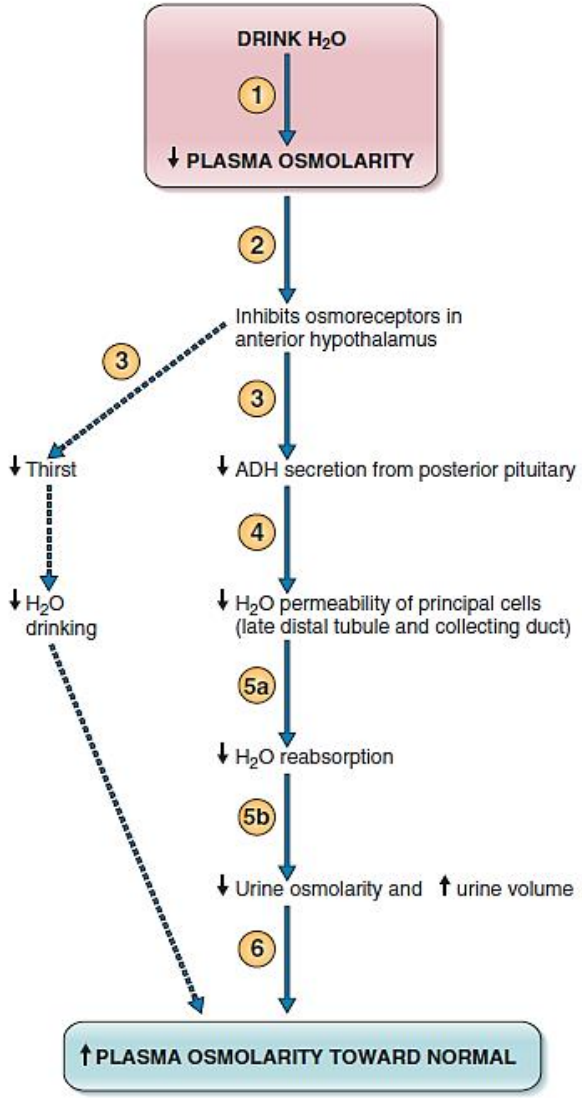
→ ↑ Plasma osmolarity
→ Stimulation of hypothalamic osmoreceptors
→ ↑ Thirst + ↑ ADH release
→ ↑ Collecting duct water permeability
→ ↑ Water reabsorption
→ ↑ Urine osmolarity + ↓ Urine volume
→ Plasma osmolarity returns toward normal



Water Drinking: The Same Feedback Loop in Reverse

- Water intake:

  → ↓ Plasma osmolarity
  → Inhibition of hypothalamic osmoreceptors
  → ↓ Thirst + ↓ ADH
  → ↓ Collecting duct water permeability
  → ↓ Water reabsorption
  → Dilute, high-volume urine
  → Plasma osmolarity returns toward normal



Osmolarity Control vs. Volume Control

Osmolarity Regulation:

- Sensor: Hypothalamic osmoreceptors
- Controlled variable: Plasma osmolarity
- Major effectors: ADH and thirst
- Renal response: Variable water excretion

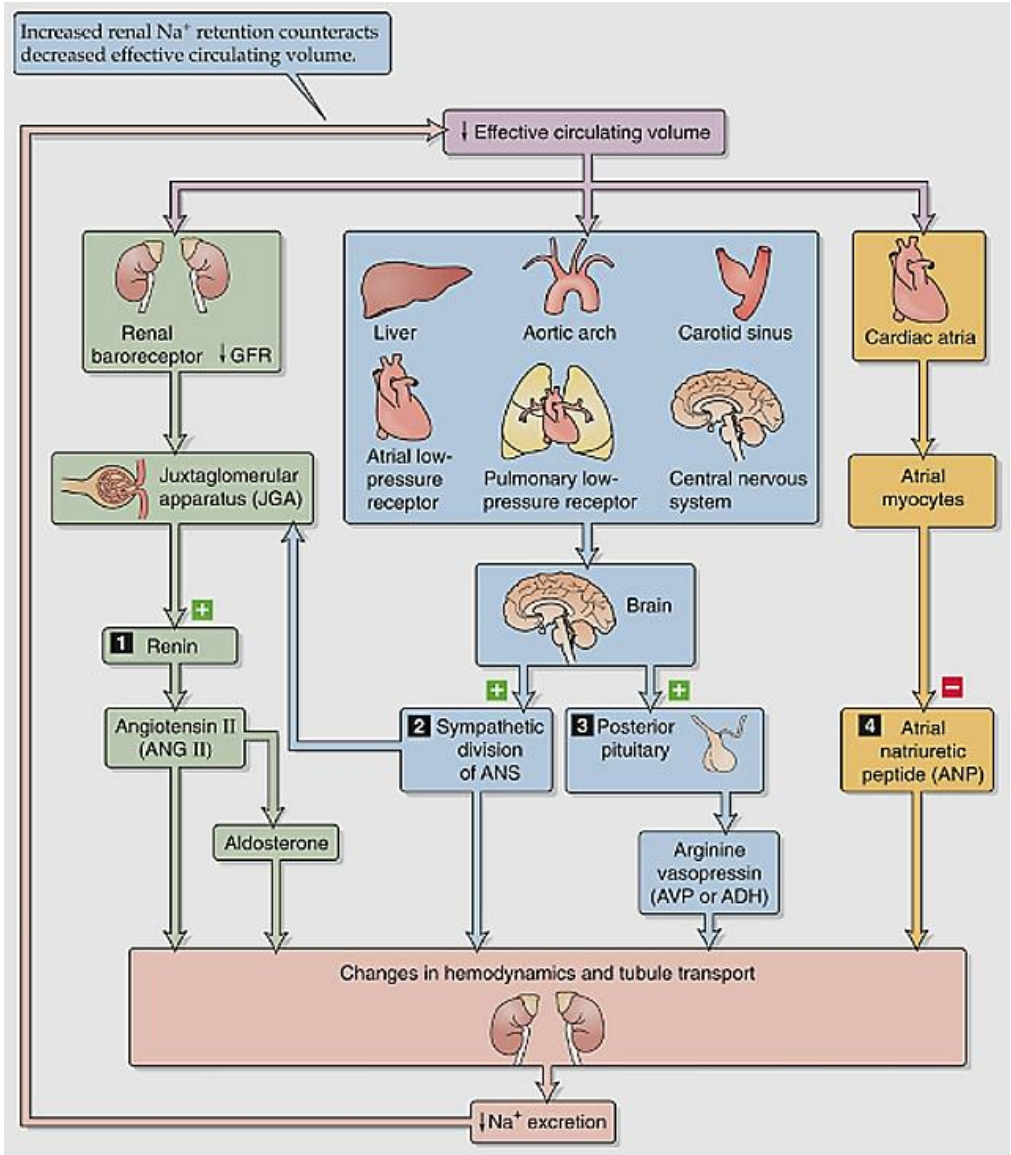
ECF Volume / EABV Regulation:

- Sensors: Arterial and cardiopulmonary pressure-volume receptors
- Controlled variable: Effective arterial perfusion
- Major effectors: Sympathetic nerves, RAAS, ANP
- **Significant volume depletion also stimulates ADH non-osmotically**

Key Principle:

- Severe volume depletion can override osmotic suppression of ADH.

Regulation of ECF Volume / EABV Regulation



Interim Summary 1

- **Target Variable Distinction:** Water balance controls plasma osmolarity/tonicity; Na^+ balance controls Extracellular Fluid (ECF) volume and Effective Arterial Blood Volume (EABV).

- **Anterior Hypothalamic Osmoreceptors:** Exquisitely sensitive (<1% osmolarity change). Hyperosmolality triggers thirst and posterior pituitary ADH release.

- **Systemic Baroreceptors:** Carotid sinus, aortic arch, and renal afferent arterioles sense arterial fullness. Hypoperfusion recruits renal sympathetic nerves and RAAS.

- **Volume Overrides Osmolality:** Severe hypovolemia (>10% volume contraction) triggers non-osmotic ADH release despite low plasma osmolarity to defend perfusion pressure.

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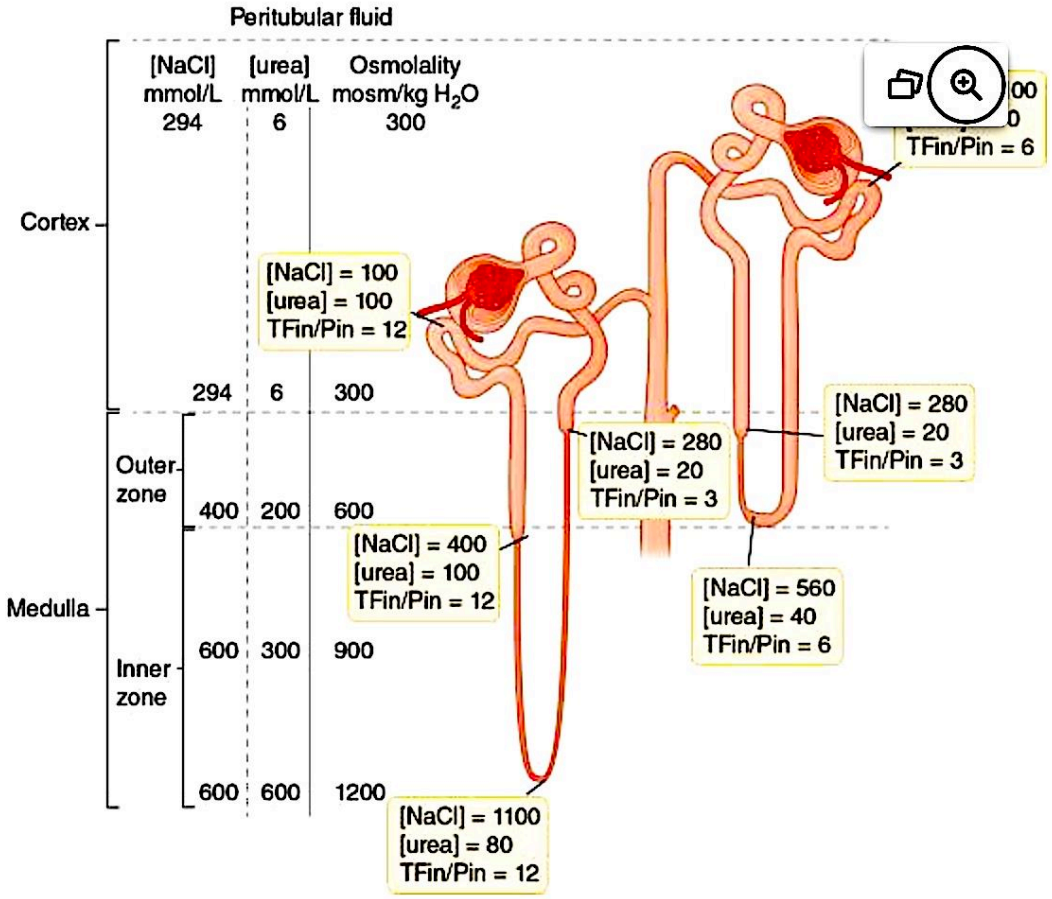
The Corticomedullary Osmotic Gradient

Cortex: ≈ 300 mOsm/L

Inner medulla / papilla: up to ≈ 1200 mOsm/L

The gradient depends on three coordinated processes:

- Loop of Henle
  - Countercurrent multiplication
  - Establishes the NaCl gradient
- Inner medullary collecting duct
  - Urea recycling
  - Strengthens the inner medullary gradient
- Vasa recta (Not shown here)
  - Countercurrent exchange
  - Preserves the gradient



Countercurrent Multiplication: The “Single Effect”

Thick Ascending Limb:
- Reabsorbs Na^+, K^+, and 2 Cl^- through NKCC2
- Essentially impermeable to water
- Tubular fluid becomes dilute
- Medullary interstitium becomes more concentrated

Descending Limb:
- Highly permeable to water
- Much less effective NaCl transport
- Water exits toward the hyperosmotic interstitium
- Tubular fluid becomes concentrated

Single Effect:
NaCl leaves ascending limb while water leaves descending limb.

Countercurrent Multiplication: Single Effect + Flow

Step 1: Single Effect
- TAL reabsorbs NaCl without water.
- Descending limb loses water.

Step 2: Tubular Flow
- New ≈ 300 mOsm fluid enters from the PCT.
- Concentrated descending-limb fluid is displaced deeper into the medulla.

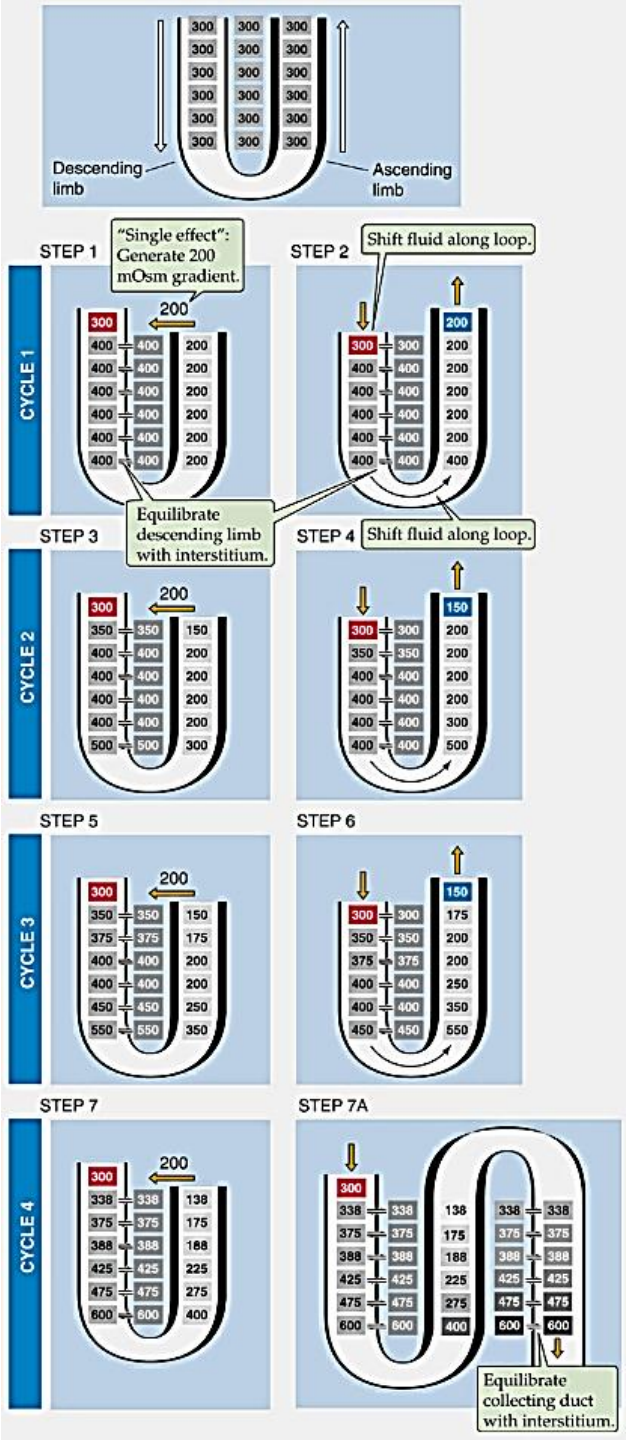
Repeated: Single Effect + Flow + Single Effect + Flow

Result: Small transverse gradients are multiplied into a large longitudinal corticomedullary gradient.

What does the countercurrent multiplier system accomplish?

It increases the concentration of renal interstitial fluid **from 300 mOsm in the cortex to 1,200 mOsm in the inner medulla**. This hypertonicity gradient in the renal medulla is critical because it **serves as the driving force for water reabsorption across the walls of the collecting ducts**, allowing the fluid in the collecting duct to equilibrate osmotically with the hyperosmotic interstitium, **producing a concentrated urine**.

*Single effect = NaCl reabsorption via the Na^+-K^+-$2Cl^-$ cotransporter; ADH increases activity of this transporter → ↑single effect



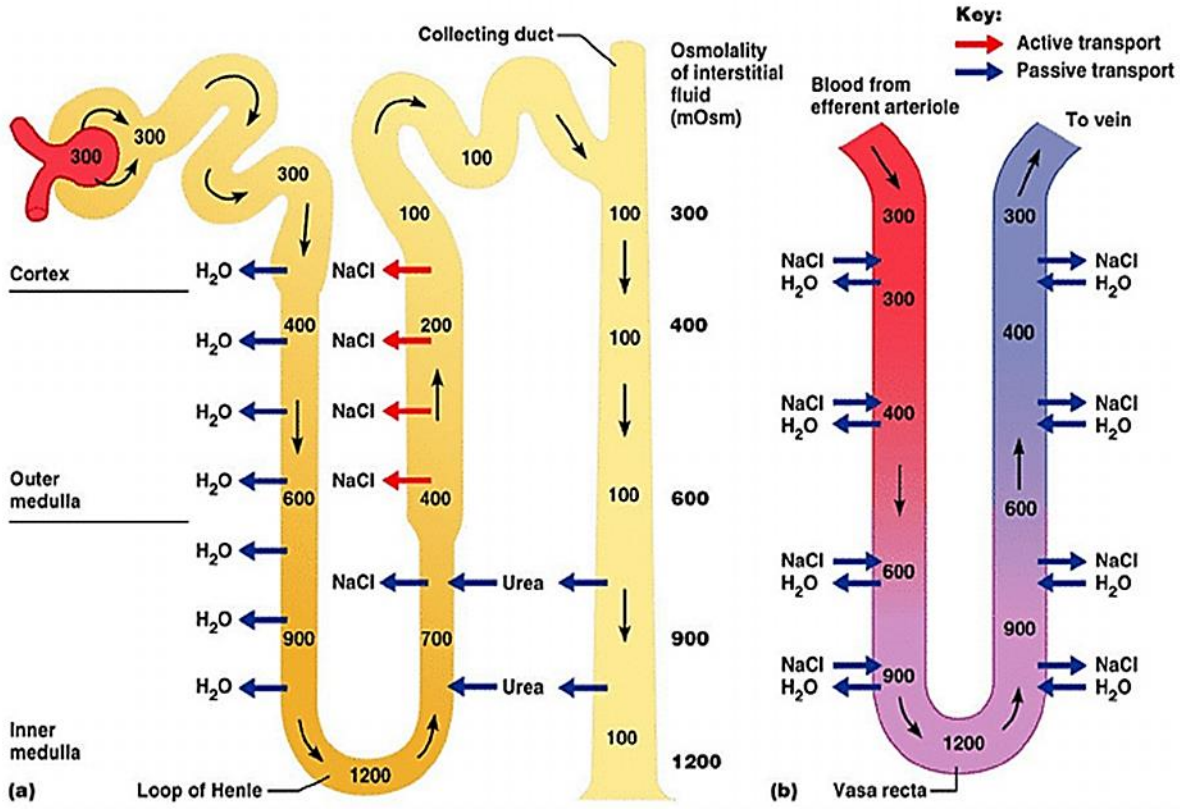
Countercurrent Multiplication

Net Effect

- Creation of a hyperosmotic interstitium
- Creation of a vertical gradient

Clinical Application

- Loop Diuretics: disrupts medullary hyperosmotic environments
- Affects urine concentrating ability



Vasa Recta: Countercurrent Exchange Preserves the Gradient

Descending Vasa Recta:
- Water leaves blood.
- NaCl and urea enter blood.
- Blood becomes progressively hyperosmotic.

Ascending Vasa Recta:
- Water enters blood.
- NaCl and urea leave blood.
- Blood becomes progressively less concentrated.

Key Features:
- Hairpin arrangement
- Slow medullary blood flow
- Passive exchange

Countercurrent exchange PRESERVES the gradient; it does not generate it.

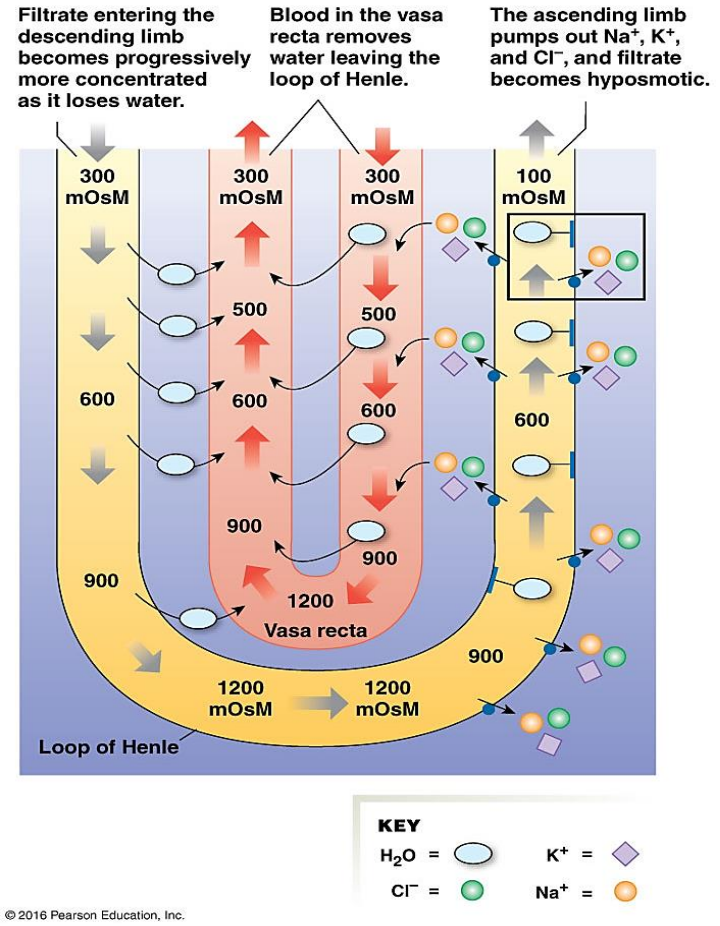
Structure –Function Relationship: Countercurrent Exchanger

- Juxtaglomerular nephrons and vasa recta run in opposite direction

- This anatomical arrangement allows effective exchange of material between tubule and vasa recta and verse versa: enhances ability to concentrate urine

- Hairpin loop arrangement of the vasa recta permits passive countercurrent exchange, minimizing washout of the corticomedullary osmotic gradient

- The countercurrent exchange system helps maintain a hyperosmotic medullary interstitium

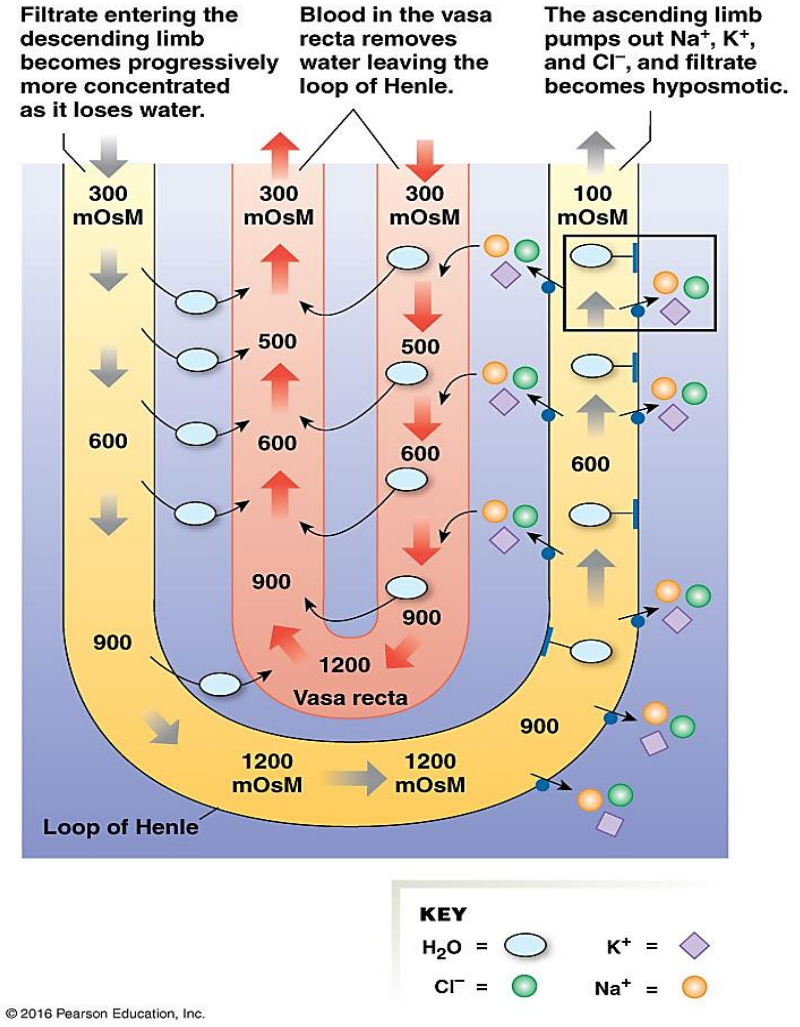


Countercurrent Exchange in Vasa Recta Maintains Hyperosmolarity of Renal Medulla

- **Two special features of renal medullary blood flow help preserve hyperosmolarity**

- **Low medullary blood flow**
- Less than 5% of total renal blood flow

- **Vasa recta serve as countercurrent exchanger**
  - **Descending** vasa recta gain solutes and lose water
  - **Ascending** lose solute and gain water
  - **Descending and ascending vessels exchange solutes and water**



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Urea Recycling: Deepening the Gradient

With ADH:

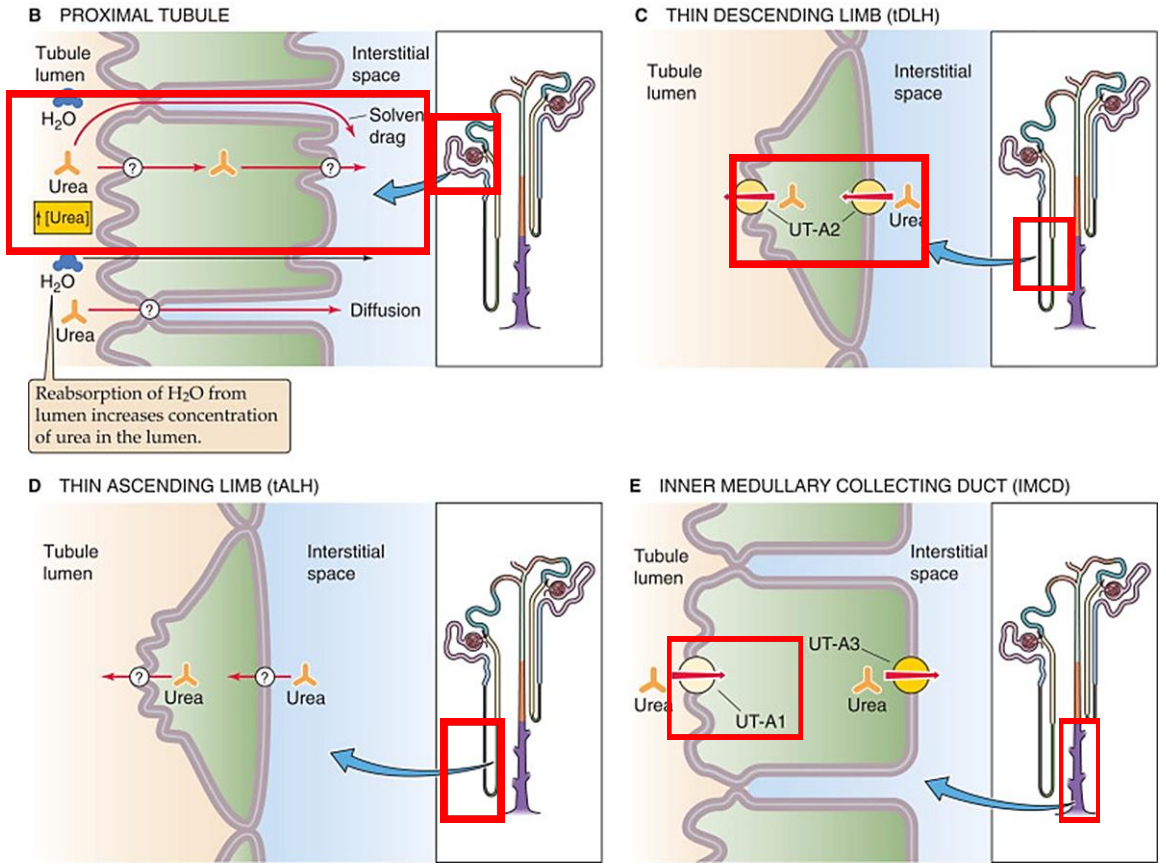
Cortical / Outer Medullary Collecting Duct:
- ↑ Water permeability
- Relatively low urea permeability
- Water leaves → tubular [urea] rises

Inner Medullary Collecting Duct:
- ↑ Water permeability
- ↑ Urea permeability
- Urea diffuses into the inner medullary interstitium

Result:
- Urea contributes to inner medullary hyperosmolarity.
- The corticomedullary gradient becomes larger during antidiuresis.

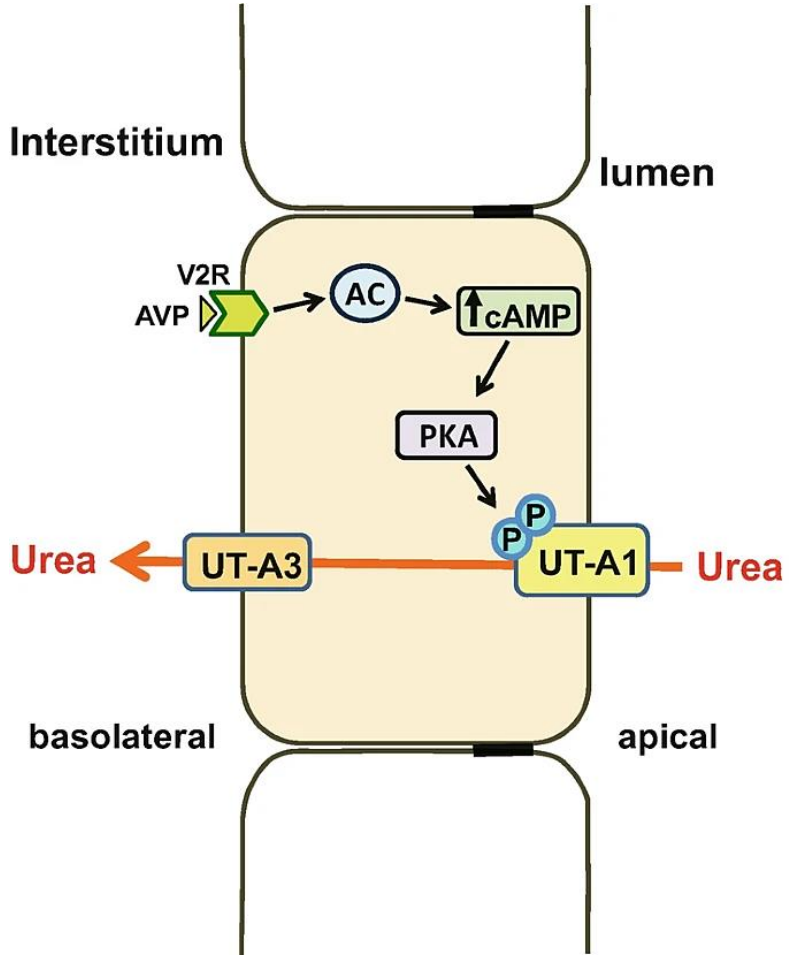
Urea Transport

- PCT transport paracellular or transcellular
- 50% reabsorbed by paracellular diffusion and solvent drag
- Descending and thin ascending limb: urea secretion occurs via UT-A2
- **Inner medullary collecting ducts (IMCD) reabsorption occurs through UT-A1 (luminal side) & UT-A3 basolateral**
- **ADH directly upregulate UT-A1 and significantly increases [urea] as water is reabsorbed**
- 70% reabsorbed in IMCD
- 40% is excreted



Urea Recycling Is Enhanced by ADH

- ADH results in phosphorylation of UT-A1 and increased luminal uptake of urea in **inner medullary collecting duct** (IMCD)
- **Water reabsorption promoted by ADH** causes luminal **concentrations of urea to increase**
- The two factors result in about 55% **urea** uptake which **accounts for ~ 50% of the medullary interstitial hyperosmolarity**
- Thus, **ADH increases both water and urea permeability**



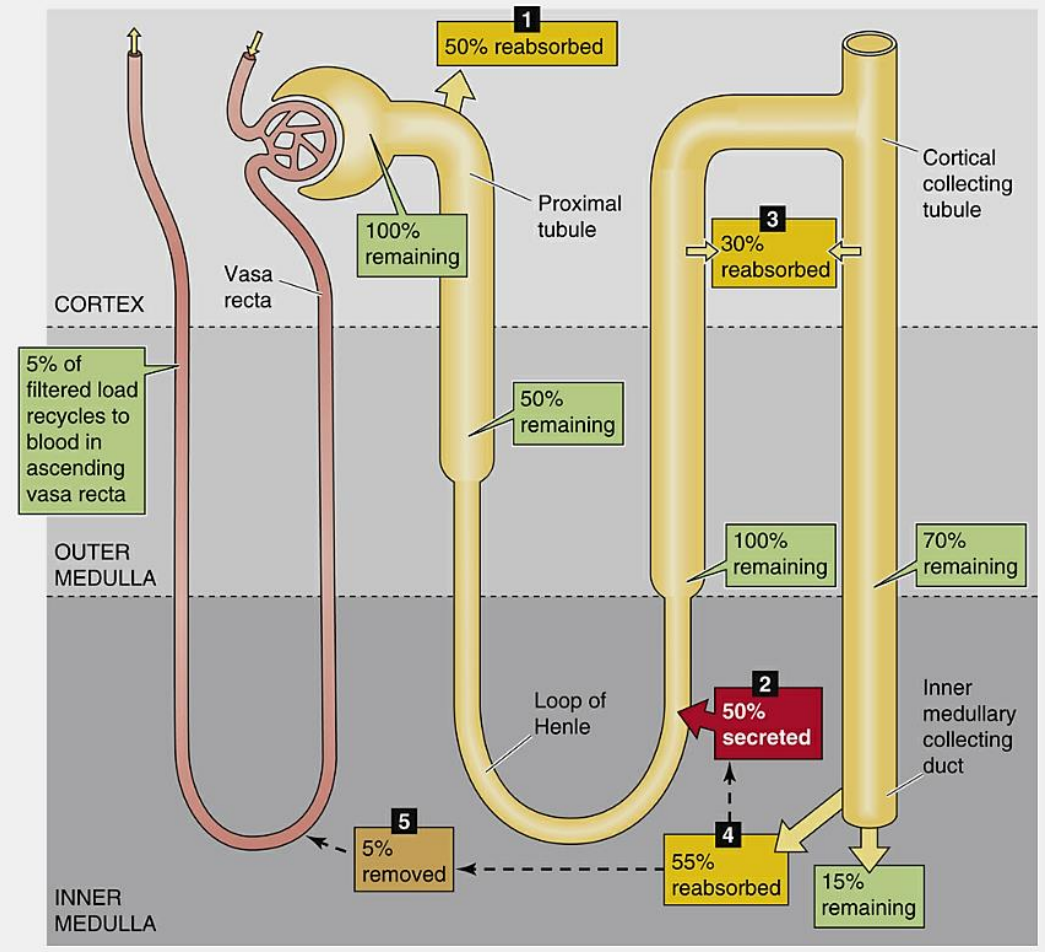
Urea Recycling

Urea contributes to the high osmolality of the medullary interstitium

- Promotes water reuptake
- Makes it possible to produce hyperosmotic urine

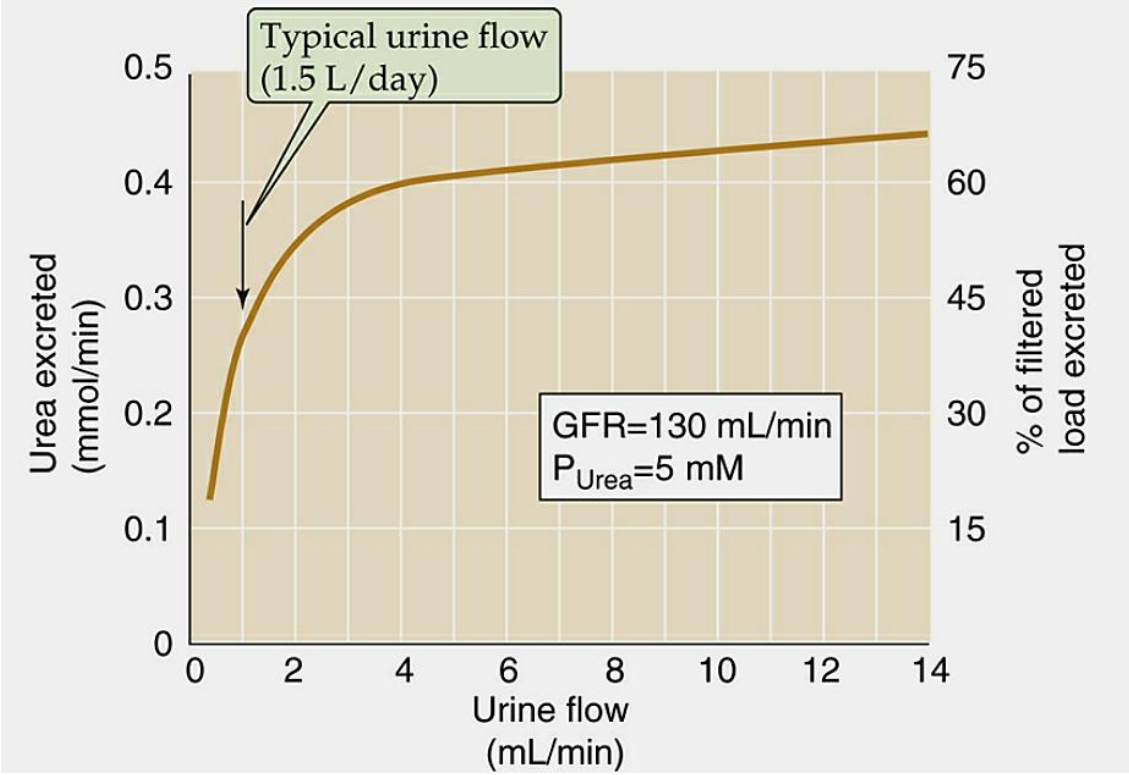
Urea Transporters (UT) promote the reuptake of urea from the lumen of **inner medullary collecting ducts (IMCDs)**

- Dependent on ADH (vasopressin)



Urea Excretion: Effect of Urine Flow

- **<u>Low urine flow</u>**: tubule reabsorbs water and urea → kidneys excrete small amount of urea (15% of filtered urea)
- **<u>High urine flow</u>**: tubule reabsorb less water and urea → kidneys can excrete up to 70%
- During progressive renal disease, GFR declines → low urine flow and urea retention **increase in BUN**
- **[BUN]/[creatinine] ratio** increase can indicate volume depletion
  - Urea excretion can decrease out of proportion with GFR (GFR normal if creatinine is normal), but urea excretion low due to decreased urine flow



Interim Summary 2

- **Corticomedullary Gradient:** Rises progressively from 300 mOsm/kg at cortex to 1200 mOsm/kg at papillary tip.

- **Countercurrent Multiplication (Loop of Henle):**
  - *Single Effect:* Thick ascending limb (TAL) actively reabsorbs NaCl via NKCC2 without water; descending limb (tDLH) passively loses water to hypertonic interstitium via AQP1.
  - *Tubular Flow:* Continuous fluid movement multiplies 200 mOsm/kg horizontal gradient into large vertical gradient.

- **Urea Recycling (Inner Medulla):** ADH upregulates UT-A1/UT-A3 in inner medullary collecting ducts (IMCD). Reabsorbed urea supplies 40–50% of papillary hypertonicity.

- **Countercurrent Exchange (Vasa Recta):** Hairpin capillary geometry and low medullary blood flow passively equilibrate solutes/water; preserves gradient without solute washout.

- **Clinical Disruption:** Loop diuretics block NKCC2, collapsing medullary gradient and abolishing both concentrating and diluting abilities.

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ADH: Three Renal Actions

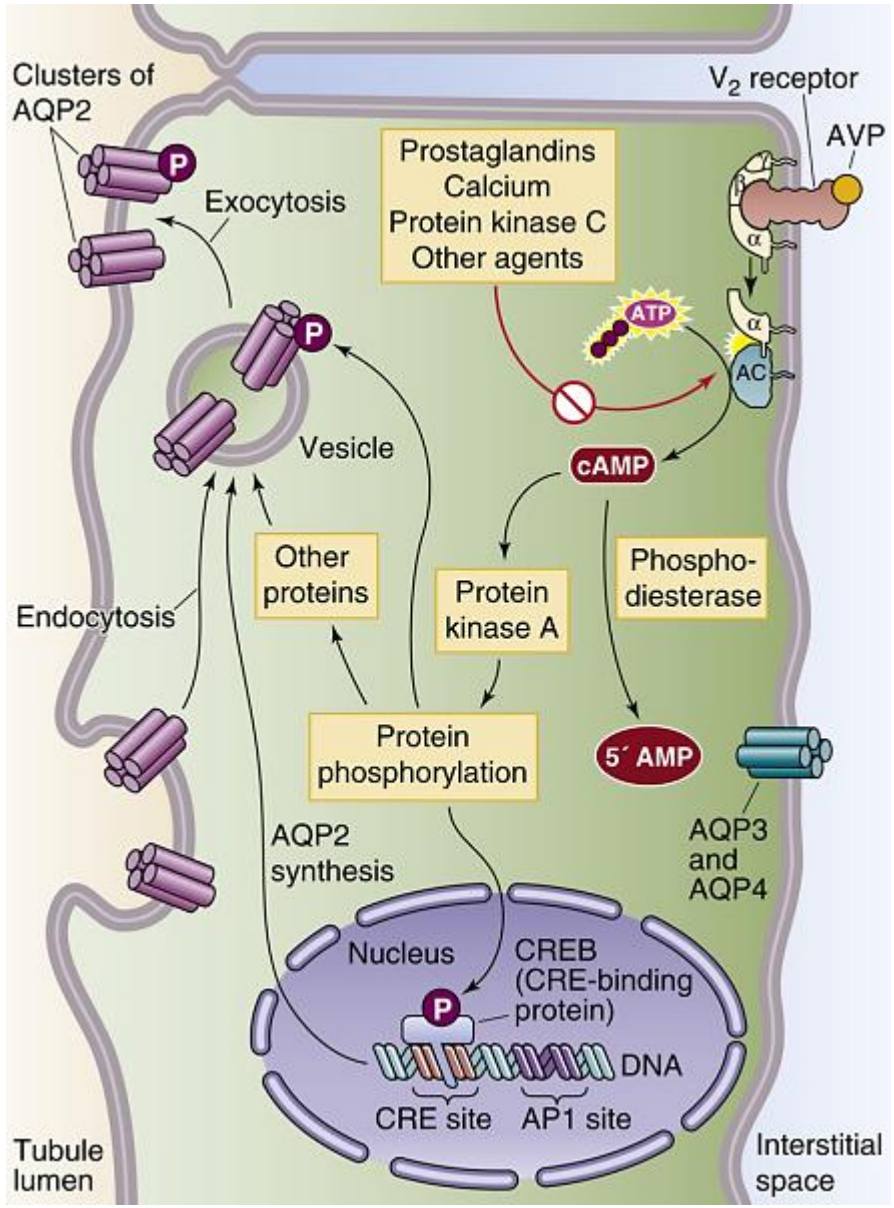
ADH:

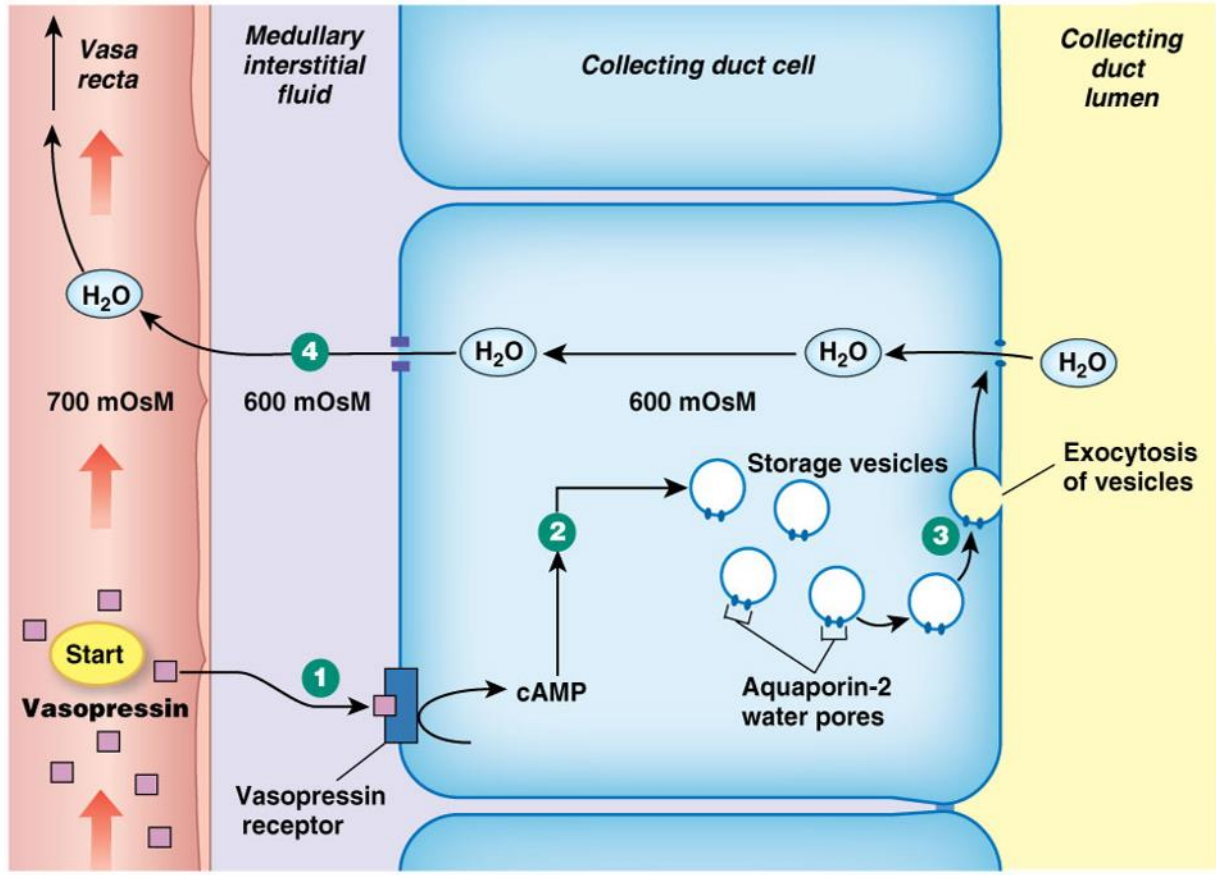
1. **Increase water permeability**
   - Late distal tubule and collecting duct principal cells
   - AQP2 insertion
2. **Increase NKCC2 activity**
   - Thick ascending limb
   - Enhances countercurrent multiplication
3. **Increase urea permeability**
   - Inner medullary collecting duct
   - Enhances urea recycling

Net Effect: Increase ability to conserve water and produce concentrated urine

ADH Signaling in the Principal Cell

1. ADH binds the basolateral V_2 receptor.
2. The V_2 receptor activates Gs protein.
3. Gs activates adenylyl cyclase.
4. Adenylyl cyclase increases intracellular cAMP.
5. cAMP activates protein kinase A (PKA).
6. PKA-dependent phosphorylation promotes trafficking of AQP2-containing vesicles.
7. AQP2 channels are inserted into the apical membrane.
8. Apical water permeability increases.
9. Water enters through AQP2 and exits basolaterally through AQP3 and AQP4.


Clusters of AQP2
V_2 receptor
AVP
Prostaglandins
Calcium
Protein kinase C
Other agents
Exocytosis
ATP
AC
Vesicle
cAMP
Other proteins
Protein kinase A
Phospho-diesterase
Endocytosis
Protein phosphorylation
5´ AMP
AQP2 synthesis
AQP3 and AQP4
Nucleus
CREB (CRE-binding protein)
DNA
CRE site
AP1 site
Tubule lumen
Interstitial space


Vasa recta
Medullary interstitial fluid
Collecting duct cell
Collecting duct lumen
H_2O
700 mOsM
600 mOsM
600 mOsM
Storage vesicles
Exocytosis of vesicles
Start
Vasopressin
cAMP
Aquaporin-2 water pores
Vasopressin receptor

Interim Summary 3

- **Triad of ADH Renal Actions:**
  - Increases water permeability in late distal tubule and collecting duct principal cells.
  - Stimulates TAL NKCC2 cotransport, augmenting single effect.
  - Upregulates IMCD UT-A1 transporters, enhancing urea recycling.

- **Principal Cell Signaling Cascade:**
  - Basolateral V_2 receptor binding → Gs activation → Adenylyl cyclase stimulation → ↑ intracellular cAMP → Protein Kinase A (PKA) activation.

- **Aquaporin Dynamics:**
  - ***Apical Insertion:*** PKA phosphorylation directs exocytic shuttling and insertion of AQP2 storage vesicles into luminal membrane.
  - ***Basolateral Efflux:*** Water exits into interstitium through constitutive, unregulated AQP3 and AQP4 channels.
  - *Signal Termination:* Absence of ADH induces endocytosis of AQP2 back into intracellular vesicles.

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High ADH: Production of Concentrated Urine

PCT:
- ≈300 mOsm/L
- Isosmotic reabsorption

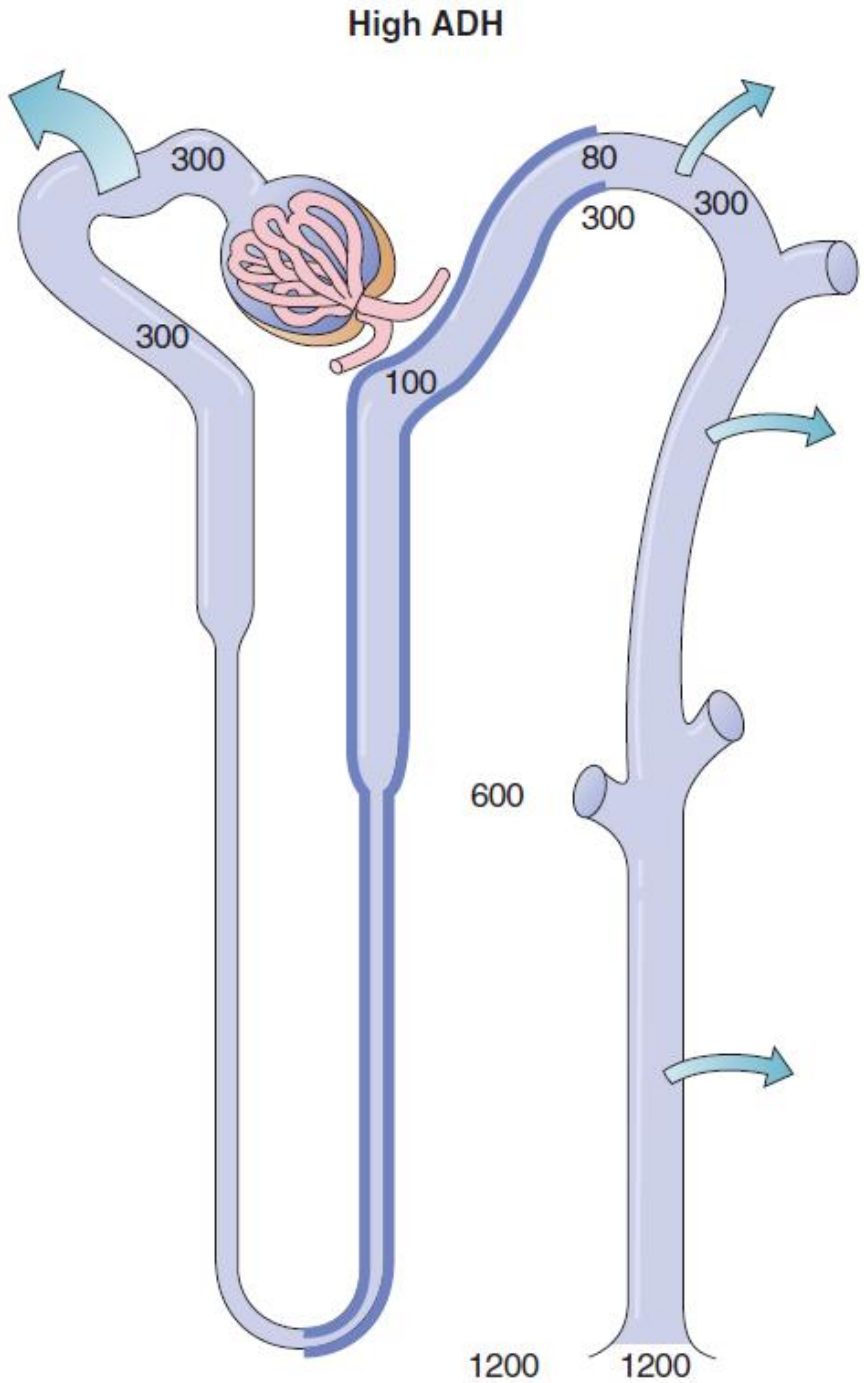
Descending Limb:
- Water exits
- Tubular fluid becomes concentrated

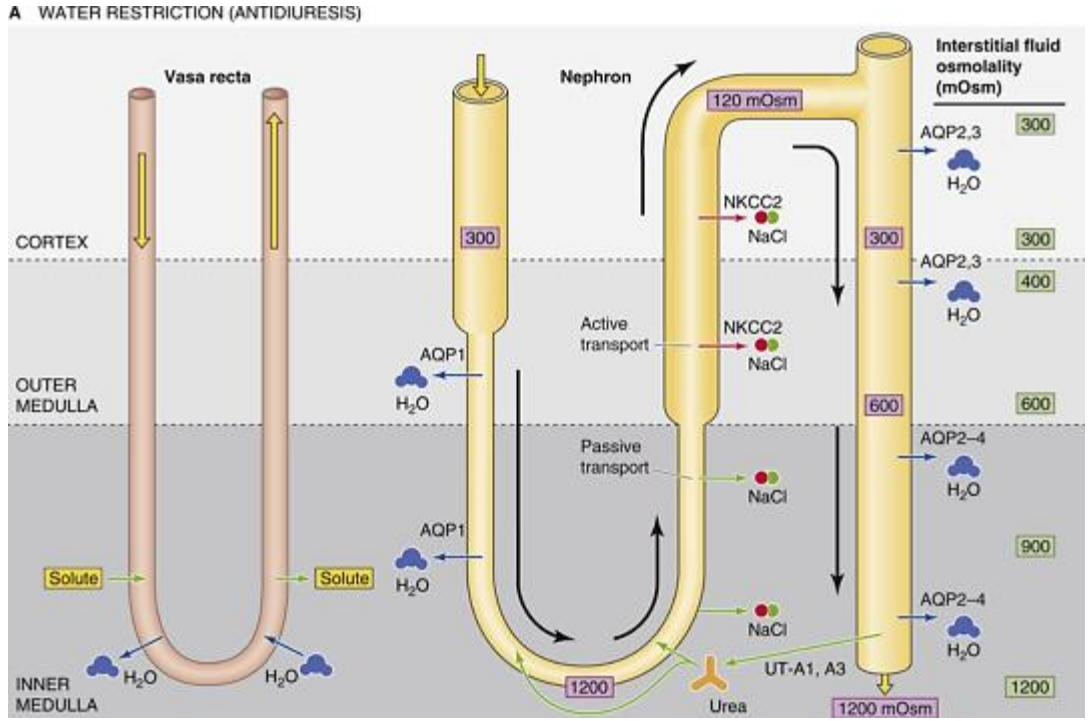
TAL:
- NaCl exits without water
- Tubular fluid becomes dilute
- ≈100 mOsm/L

Early DCT:
- Further NaCl reabsorption without water
- ≈80 mOsm/L

Late DCT / Collecting Duct + ADH:
- High water permeability
- Water follows corticomedullary osmotic gradient
- Tubular fluid becomes progressively concentrated

Final Urine:
- Low volume
- Hyperosmotic
- Can approach ≈1200 mOsm/L


High ADH
300
300
100
80
300
300
600
1200
1200


A WATER RESTRICTION (ANTIDIURESIS)
Vasa recta
Nephron
Interstitial fluid osmolality (mOsm)
120 mOsm
CORTEX
OUTER MEDULLA
INNER MEDULLA
AQP2,3
NKCC2
NaCl
Active transport
AQP1
Passive transport
AQP2–4
UT-A1, A3
Urea
Solute
1200 mOsm
300
400
600
900
1200

Low or Absent ADH: Production of Dilute Urine

PCT:

- Water handling unchanged
- Tubular fluid remains ≈300 mOsm/L

TAL + Early DCT:

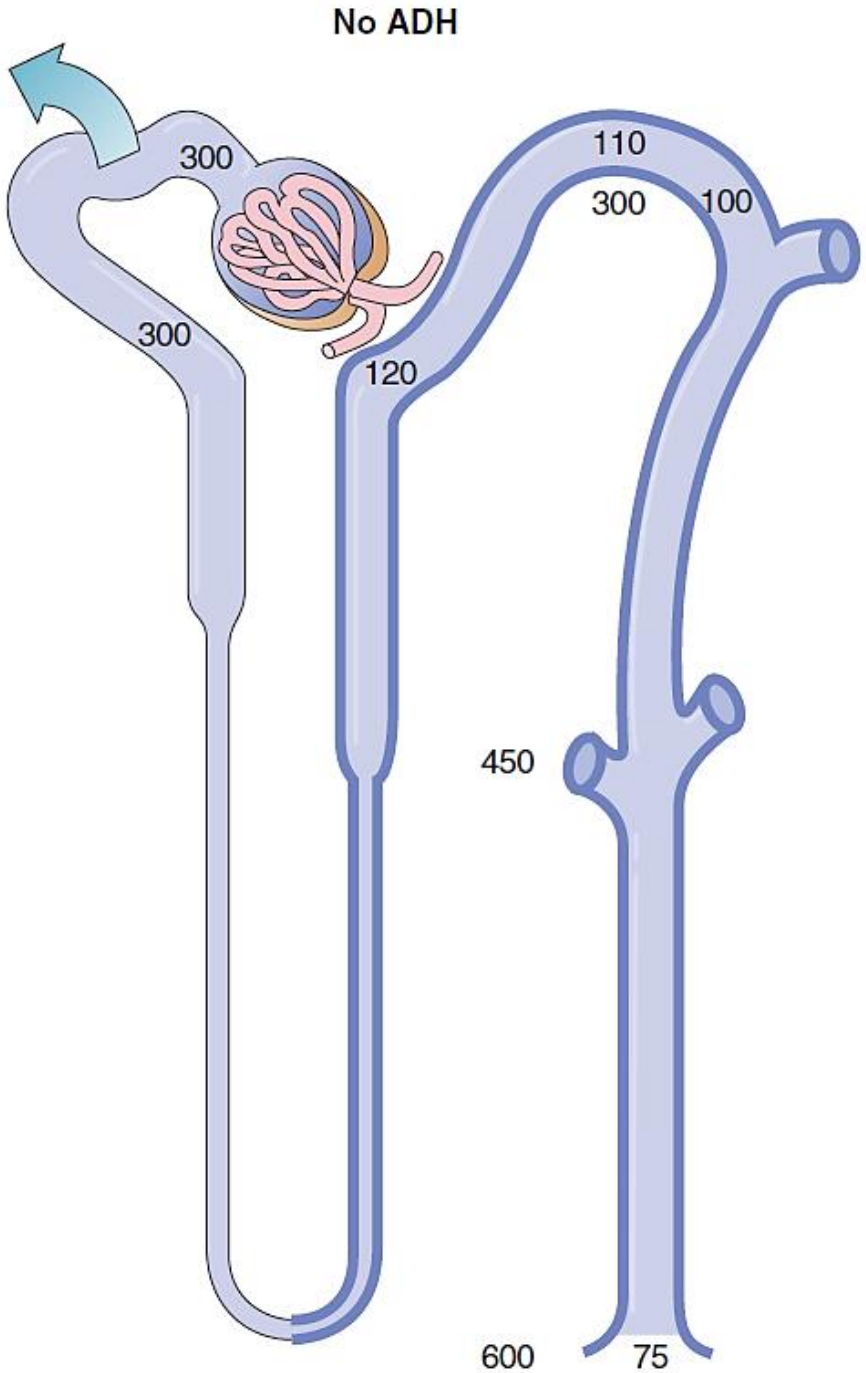
- NaCl reabsorbed without water
- Tubular fluid becomes progressively dilute

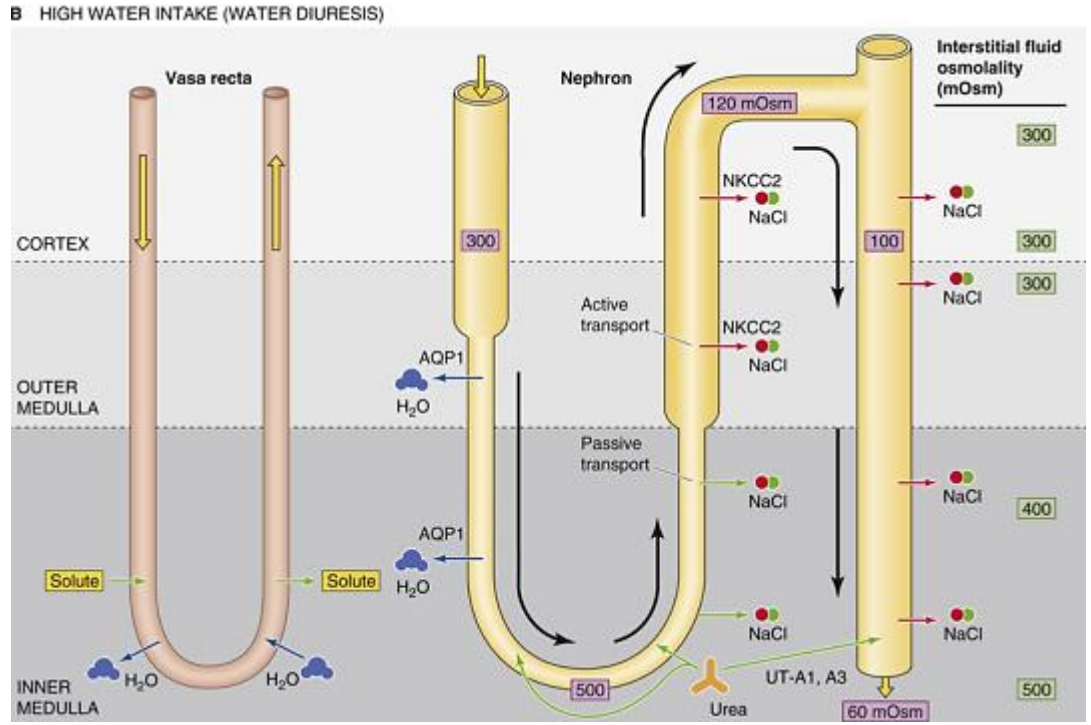
Late DCT / Collecting Duct:

- Low water permeability
- No osmotic equilibration with medullary interstitium
- Additional solute can be reabsorbed while water remains

Final Urine:

- High volume
- Hyposmotic
- Can fall to approximately 50–100 mOsm/L


No ADH
300
110
300
100
300
120
450
600
75


B HIGH WATER INTAKE (WATER DIURESIS)
Vasa recta
Nephron
Interstitial fluid osmolality (mOsm)
120 mOsm
CORTEX
OUTER MEDULLA
INNER MEDULLA
300
100
NKCC2
NaCl
Active transport
AQP1
H2O
Passive transport
Solute
500
UT-A1, A3
Urea
60 mOsm
400

Summary: High ADH vs. Low ADH

HIGH ADH:

- Collecting duct water permeability ↑
- AQP2 inserted
- Urea recycling ↑
- Medullary gradient strengthened
- Urine volume ↓
- Urine osmolarity ↑
- Free-water clearance negative

LOW / INEFFECTIVE ADH:

- Collecting duct water permeability ↓
- AQP2 insertion ↓
- Urea recycling ↓
- Urine volume ↑
- Urine osmolarity ↓
- Free-water clearance positive

Renal Handling of Water

Key principle: Regardless of the volume of water excreted, kidneys must excrete ~600 milliosmoles/day, i.e. product of urine osmolality and urine output is approximately constant

- 600 mOsm/day = U_{osm} x $\dot{V}$ of urine output/day

- Normal 70kg Person excretes ~ 600 mOsm/day of solute

- Osmoles excreted/day = U_{osm} x $\dot{V}$ of urine output/day

Renal Handling of Water

If you excrete 1500 mL of urine each day, what is the osmolality of the urine?

- Osmoles excreted/day = U_{osm} x $\dot{V}$ of urine output/day
- 600 mOsm/day = U_{osm} x 1.5 L/day
- 600 mOsm/day/1.5 L/day = **400 mOsm/L**

Renal Handling of Water

With high water intake, urine osmolality can be as low as ~ 40 mOsm

- If: Osmoles excreted/day = U_{osm} x $\dot{V}$ of urine output/day and 600 mOsm/day must be excreted, what is urine flow rate, $\dot{V}$?
- 600 mOsm/day = 40 mOsm x $\dot{V}$
- **$\dot{V}$ = 15 L/day**

Obligatory Urine Volume

- The maximal concentrating ability of the kidney determines the minimal volume of urine that can be excreted each day

- Maximal urine concentration = 1200 mOsm/L

- Osmolar excretion $= U_{\text{osm}} \times \dot{V}$

- Obligatory urine volume = **600 mOsm/day**
  1200 mOsm/L
  = 0.5 L/day

Concentration & Dilution of Urine

- Kidneys will dilute or concentrate urine as needed to regulate water balance:
  - **Increase urine volume; decrease urine osmolality**
  - **Decrease urine volume; increase urine osmolality**
- **Renal failure** reduces both the **concentrating** and **diluting** ability

Free-Water Clearance: What Does It Mean?

Free water = water in excess of the amount required to excrete urinary solute at plasma osmolarity.

Diluting Segments:

- Thick ascending limb
- Early distal tubule
- Reabsorb solute without water
- Generate solute-free water

Osmolar Clearance:

$$C_{osm} = (U_{osm} \times \dot{V}) / P_{osm}$$

Free-Water Clearance:

$$C_{H_2O} = \dot{V} - C_{osm}$$

Interpretation:

- $C_{H_2O} > 0$ → free water is being excreted (positive)
- $C_{H_2O} < 0$ → free water is being reabsorbed (negative)
- $C_{H_2O} = 0$ → urine is isosmotic with plasma

Osmolar Clearance & Free Water Clearance

- Osmolar Clearance (C_{osm}) = Volume of plasma cleared of solutes/time unit

- **$C_{osm} = (U_{osm}\ \dot{V})/\ P_{osm}$**

Osmolar Clearance Example

Given that plasma osmolarity is 300 mOsm/L, urine osmolarity is 600 mOsm/L and the urine flow rate is 1 ml/min (0.001 L/min) calculate the osmolar clearance

- $C_{osm} = (U_{osm} \times \dot{V}) / P_{osm}$

  $= \frac{600\ mOsm/L * 1 ml/min}{300\ mOsm/L}$

  $= \underline{\mathbf{2\ ml/min}}$

This means

2 ml of plasma is being completely cleared of solutes each minute

Free Water Clearance

- Free water Clearance (C_{H2O}) = Difference between urine flow rate ($\dot{V}$) and Osmolar clearance (C_{osm})

- **$C_{H2O} = \dot{V} - C_{Osm}$**

- **$\dot{V} = C_{osm} + C_{H2O}$**

Free Water Clearance Calculation

Determine free water clearance given that urine osmolality is 100 mOsm/L, urine flow rate is 10 mL/min and plasma osmolality is 300 mOsm/L

Step 1. Determine Osmolar clearance

- $C_{osm} = U_{osm} \times \dot{V} / P_{osm}$
  $= [100 \text{ mOsm/L} \times 10 \text{ mL/min}]/300 \text{ mOsm/L}$
  $= 3.3 \text{ mL/min}$

Step 2. Subtract Osmolar clearance from urine flow rate

- $C_{H2O} = \dot{V} - C_{Osm}$
  $C_{H2O} = 10 - 3.3$
  $C_{H2O} = +6.7 \text{ mL/min}$

A man has a urine flow rate of 10 mL/min, a urine osmolarity of 100 mOsm/L, and a plasma osmolarity of 290 mOsm/L. What is his free-water clearance, and what is its significance?

Given:

- Urine flow rate, **$\dot{V}$ = 10 mL/min**
- Urine osmolarity, **U_{osm} = 100 mOsm/L**
- Plasma osmolarity, **P_{osm} = 290 mOsm/L**

Step 1: Calculate Osmolar Clearance

- C_{osm} = (100 mOsm/L× 10) / 290
- C_{osm} = 3.45 mL/min

Step 2: Calculate Free-Water Clearance

- C_{H_2O} = (10 − 3.45) mL/min
- C_{H_2O} = +6.55 mL/min

$$C_{H_2O} = \dot{V} - C_{osm}$$

$$= \dot{V} - \frac{[U]_{osm} \times \dot{V}}{[P]_{osm}}$$

$$= 10\text{mL/min} - \frac{100\text{mOsm/L} \times 10\text{mL/min}}{290\text{mOsm/L}}$$

$$= 10\text{mL/min} - 3.45\text{mL/min}$$

$$= +6.55\text{mL/min}$$

Interpretation: Positive C_{H_2O} = net excretion of solute-free water.

Means that free water is being excreted. The solute-free water generated in the thick ascending limb and early distal tubule is not reabsorbed by the collecting ducts, but it is excreted. This situation occurs when circulating ADH levels are low, as in water drinking or central diabetes insipidus (or if ADH is ineffective, as in nephrogenic diabetes insipidus).

Free Water Clearance

- C_{H2O} can be as **high as + 18 L/Day** under maximally diluting conditions
- C_{H2O} can be as **low as - 1.5 L/Day** under maximally concentrating conditions
- The wide range of C_{H2O} represents the kidneys' attempt to stabilize the osmolality of the ECF in the face of changing loads of solutes and water

If $U_{osm} = P_{osm}$

- Then Osmolar Clearance (C_{osm}) = Urine flow rate ($\dot{V}$)
- Free water Clearance $C_{H2O} = 0$

Positive Free Water Clearance (Hypoosmotic Urine)

- Dilute Urine (Urine osmolarity U_{osm} < Plasma osmolarity P_{osm})
- $\dot{V} > C_{osm}$
  - C_{H2O} is positive; free water movement
  - **Produced in high water intake**

- If volume of urine output is 20 L/day and the C_{osm} is 2 L/day
- $C_{H2O} = \dot{V} - C_{Osm}$
  - C_{H20} = 20 L/day – 2 L/day
  - **C_{H2O} = <u>18 L/day</u>**

Positive Free Water (Hypoosmotic Urine)

Occurs under the following conditions:

- Circulating **ADH** levels are **LOW**
  - High water intake
  - Central diabetes insipidus
  - Low levels of ADH
  - Nephrogenic diabetes insipidus
    - ADH is ineffective: Receptor insensitivity

Negative Free Water Clearance (Hyperosmotic Urine)

Concentrated urine ($U_{osm} > P_{osm}$)

- $\dot{V} < C_{osm}$
  - C_{H2O} is negative
  - Produced in water deprivation (kidney is removing water from tubular fluid)
  - With restricted water intake, kidney is capable of producing U_{osm} as high as 1200mOsm/L

Obligatory urine volume is 0.5 L/day and the C_{osm} of 2 L/day

- $C_{H2O} = \dot{V} - C_{osm}$
- = [0.5 - 2]L/day
  - C_{H2O} = -1.5 L/day

Negative Free Water (Hyperosmotic Urine)

Urine osmolality > blood osmolality

Occurs under the following conditions:

- Circulating **Antidiuretic Hormone (ADH)** levels are **HIGH**
- Water deprivation
- Volume depletion
- SIADH

Disorders of Urinary Concentrating Ability

- Inappropriate secretion of ADH
- Impairment of countercurrent mechanism
  - Even in the presence of ADH urine concentration is limited by the degree of medullary hyperosmolarity
- Inability of Initial Collecting Tubules (ICT), Cortical Connecting Tubules (CCT), and Collecting Duct (CD) to respond to ADH

Interim Summary 4

- **Nephron Osmolarity Profile:**
  - *PCT:* Isosmotic reabsorption (300 mOsm/kg; $[TF/P]_{osm} = 1.0$).
  - *TAL & Early DCT:* Watertight "diluting segments" reabsorb solute without water; fluid exits hypoosmotic (80–100 mOsm/kg).
  - *Collecting Duct (+ADH):* Osmotic equilibration yields low-volume, hyperosmotic urine (up to 1200 mOsm/kg).
  - *Collecting Duct (-ADH):* Watertight tubule excretes high-volume, hypoosmotic urine (50–100 mOsm/kg).

- **Key Clearance Equations:**
  - *Osmolar Clearance:* $C_{osm} = (U_{osm} \times \dot{V}) / P_{osm}$
  - *Free Water Clearance:* $C_{H_2O} = \dot{V} - C_{osm}$

- **C_{H_2O} Clinical Interpretation:**
  - *Positive ($C_{H_2O} > 0$): Solute-free water excreted; urine hypoosmotic ($U_{osm} < P_{osm}$); seen in water drinking, central DI, nephrogenic DI.*
  - *Negative ($C_{H_2O} < 0$): Solute-free water reabsorbed; urine hyperosmotic ($U_{osm} > P_{osm}$); seen in water deprivation, SIADH.*
  - *Zero ($C_{H_2O} = 0$): Urine isosmotic with plasma ($U_{osm} = P_{osm}$); can occur when loop diuretics block solute-free water generation in TAL.*

Learning Objectives

1. Contrast the anterior hypothalamic osmoreceptor-ADH feedback system regulating body fluid osmolarity with baroreceptor-mediated pathways governing extracellular fluid volume.

2. Explain how countercurrent multiplication in the loop of Henle, urea recycling via inner medullary collecting ducts, and countercurrent exchange in the vasa recta establish and preserve the hyperosmotic corticomedullary gradient.

3. Detail the V_2 receptor-mediated cellular signaling cascade in collecting duct principal cells that directs aquaporin-2 trafficking and regulates luminal water permeability.

4. Trace tubular fluid osmolarity changes along the nephron during states of water deprivation versus water diuresis, and calculate osmolar and free-water clearance.

5. Differentiate the etiologies of central diabetes insipidus, nephrogenic diabetes insipidus, and SIADH.

Central DI, Nephrogenic DI, and SIADH

CENTRAL DI

- ADH: ↓↓
- Plasma osmolarity: ↑
- Urine osmolarity: ↓
- Urine flow: ↑
- C_{H_2O}: Positive
- DDAVP response: Strong

NEPHROGENIC DI

- ADH: ↑
- Plasma osmolarity: ↑
- Urine osmolarity: ↓
- Urine flow: ↑
- C_{H_2O} : Positive
- DDAVP response: Minimal or absent

SIADH

- ADH activity: Inappropriately ↑
- Plasma osmolarity: ↓
- Urine osmolarity: Inappropriately high
- Urine flow: ↓
- C_{H_2O}: Negative

TABLE 6.10

Antidiuretic Hormone Examples of Physiology and Pathophysiology

Example	Serum ADH	Plasma Osmolarity	Urine Osmolarity	Urine Flow Rate	Free-Water Clearance (C_{H_2O})
Water deprivation	↑	High-normal	Hyperosmotic	Low	Negative
SIADH	↑ ↑	Low (reabsorption of excess water)	Hyperosmotic	Low	Negative
Water drinking	↓	Low-normal	Hyposmotic	High	Positive
ADH deficiency (central diabetes insipidus)	↓ ↓	High (excretion of excess water)	Hyposmotic	High	Positive
ADH resistance (nephrogenic diabetes insipidus)	↑ (Stimulated by high plasma osmolarity)	High (excretion of excess water)	Hyposmotic	High	Positive

ADH, Antidiuretic hormone; *SIADH*, syndrome of inappropriate antidiuretic hormone.

Interim Summary 5

- **Central Diabetes Insipidus:**
  - *Etiology:* Impaired hypothalamic ADH synthesis or pituitary release (head trauma, pituitary surgery).
  - *Findings:* Serum ADH low/undetectable, hyperosmotic plasma, dilute polyuria (U_{osm} < 100 mOsm/kg), positive C_{H_2O}.

- **Nephrogenic Diabetes Insipidus:**
  - *Etiology:* End-organ renal resistance to ADH (mutated V_2 receptor/AQP2, lithium toxicity, hypercalcemia).
  - *Findings:* Serum ADH elevated, hyperosmotic plasma, dilute polyuria, positive C_{H_2O}.

- **SIADH (Syndrome of Inappropriate ADH):**
  - *Etiology:* Autonomous, unregulated ADH secretion (small cell lung cancer, CNS infections, drugs).
  - *Findings:* Inappropriately concentrated urine (U_{osm} > P_{osm}), negative C_{H_2O}, ECF volume expansion leading to dilutional hyponatremia.

Master Summary: How the Kidney Controls Water

SENSOR:

- Plasma osmolarity
- Hypothalamic osmoreceptors

SIGNAL:

ADH: → V_2 → Gs → cAMP → PKA

MEMBRANE RESPONSE:

- **AQP2 insertion**
- → Variable collecting duct water permeability

DRIVING FORCE: Corticomedullary osmotic gradient

GRADIENT MECHANISMS:

- Loop of Henle → creates
- Urea recycling → strengthens
- Vasa recta → preserves

FINAL OUTPUT:

High ADH

- Concentrated urine
- Negative C_{H_2O}

Low / ineffective ADH

- Dilute urine
- Positive C_{H_2O}

Can you explain WHY each of these occurs?

1. Water deprivation: High ADH; Concentrated urine

2. Water drinking: Low ADH; Dilute urine

3. Central DI: Low ADH; Dilute urine despite high plasma osmolarity

4. Nephrogenic DI: High ADH; Dilute urine despite high plasma osmolarity

5. SIADH: High ADH activity; Low plasma osmolarity + inappropriately concentrated urine

The End

Appendix Information

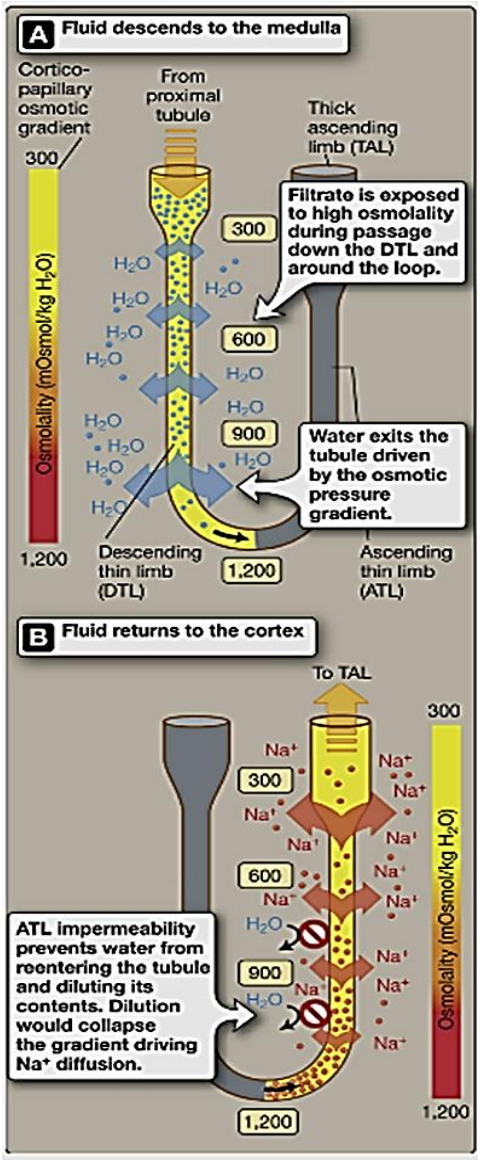
Reference materials, reading beyond the presentation, and anything else that needs to be included for completeness!

Countercurrent Multiplication

Creation of medullary hyperosmotic interstitium:

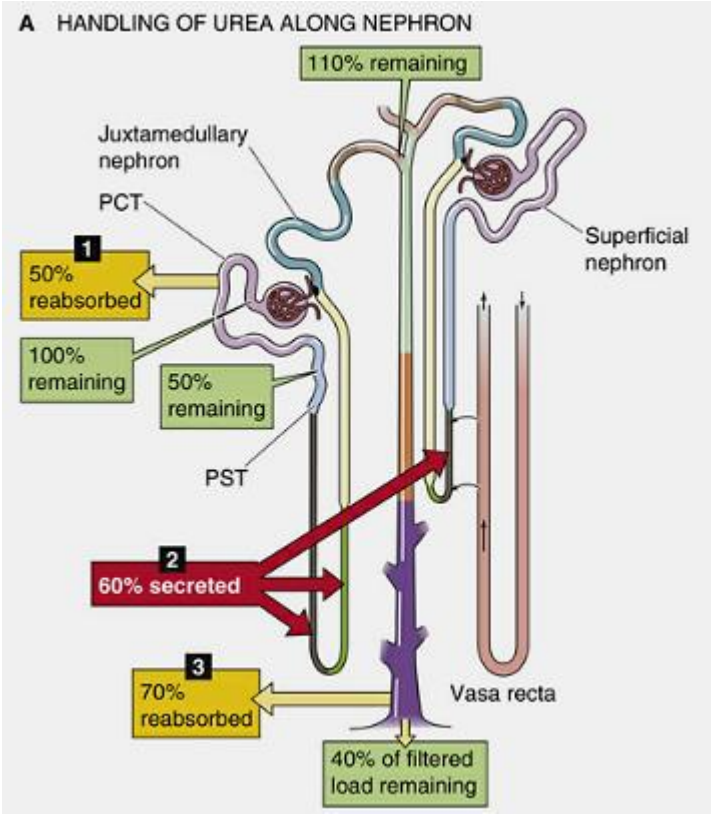
Loop of Henle Characteristics

- Active pump in the ascending limb
- Water does not follow
- Water diffuses out of the descending limb



Urea Transport

- Urea freely filtered at glomerulus, reabsorbed and some secreted
- **Half** reabsorbed in the proximal tubule
- ~60% of urea is secreted back into loop of Henle
- ~70% reabsorbed a second time in medullary collecting duct
- Results in ~40% of filtered load excreted
- The recycling of urea helps maintain interstitial medullary hyperosmotic environment



Structure	Abbreviation	Meaning	Units and/or Normal Value
Whole Kidney	C	Clearance	mL/min
	[U]	Concentration in urine	mg/mL
	[P]	Concentration in plasma	mg/mL
	$\dot{V}$	Urine flow rate	mL/min
	GFR	Glomerular filtration rate	120 mL/min
	RPF	Renal plasma flow	660 mL/min
	RBF	Renal blood flow	1200 mL/min
Single Nephron	[TF]	Concentration in tubular fluid	mg/mL
	$[TF/P]_x$	Concentration of x in tubular fluid relative to concentration of x in plasma	None
	$[TF/P]_{inulin}$	Concentration of inulin in tubular fluid relative to concentration of inulin in plasma	None
	$[TF/P]_x/[TF/P]_{inulin}$	Fraction of the filtered load remaining in tubular fluid, or fractional excretion	None